

Packet Tracer – IPv6 Neighbor Discovery

Addressing Table

Device	Interface	IPv6 Address / Prefix	Default Gateway
RTA	G0/0/0	2001:db8:acad:1::1/64	N/A
	G0/0/1	2001:db8:acad:1::1/64	N/A
PCA1	NIC	2001:db8:acad:1::A/64	fe80::1
PCA2	NIC	2001:db8:acad:1::B/64	fe80::1
PCB1	NIC	2001:db8:acad:2::A/64	fe80::1

Objectives

Part 1: IPv6 Neighbor Discovery Local Network

Part 2: IPv6 Neighbor Discovery Remote Network

Background

In order for a device to communicate with another device, the MAC address of the destination must be known. With IPv6, a process called Neighbor Discovery using NDP or ND protocol is responsible for determining the destination MAC address. You will gather PDU information in simulation mode to better understand the process. There is no Packet Tracer scoring for this activity.

Instructions

Part 1: IPv6 Neighbor Discovery Local Network

In Part 1 of this activity, you will obtain the MAC address of a destination device on the same network.

Step 1: Check the router for any neighbors that it discovered.

- Click the RTA Router. Select the CLI tab and issue the command **show ipv6 neighbors** from the privileged exec mode. If there are any entries displayed, remove them using the command **clear ipv6 neighbors**.
- Click **PCA1**, select the Desktop tab and click the **Command Prompt** icon.

Step 2: Switch to Simulation Mode to capture events.

- Click the **Simulation** button in the lower right corner of the Packet Tracer Topology window.
- Click the **Show All/None** button in the lower left part of the Simulation Panel. Make certain **Event List Filters – Visible Events** displays **None**.
- From the command prompt on **PCA1**, issue the command **ping -n 1 2001:db8:acad:1::b**. This will start the process of pinging **PCA2**.
- Click the **Play Capture Forward** button, which is displayed as an arrow pointing to the right with a vertical bar within the Play Controls box. The status bar above the Play Controls should read Captured to 150. (The exact number may vary.)

- g. Click the **Edit Filters** button. Select the IPv6 tab at the top and check the boxes for **ICMPv6** and **NDP**. Click the red X in the upper right of the Edit ACL Filters window. The captured events should now be listed. You should have approximately 12 entries in the window.

Why are ND PDUs present?

PCA1 needs to know the MAC address of PCA2 in order to send ICMPv6 ping packets to it. IPv6 ND asks for this information on the network.

- h. Click the square in the Type column for the first event, which should be **ICMPv6**.

Because the message starts with this event there is only an Outbound PDU. Under the OSI Model tab, what is the Message Type listed for ICMPv6?

ICMPv6 Echo Message Type: 128

Notice there is no Layer 2 addressing. Click the **Next Layer >>** button to get an explanation about the ND (Neighbor Discovery) process.

- i. Click the square next to the next event in the Simulation Panel. It should be at device PCA1 and the type should be NDP.

What changed in the Layer 3 addressing?

The destination address is now FF02::1:FF00:B, which is an IPv6 multicast address.

What Layer 2 addresses are shown?

The source MAC address is PCA1 MAC – 0001.427E.E8ED, and the destination MAC address is 3333.FF00.000B.

When a host does not know the MAC address of the destination, a special multicast MAC address is used by IPv6 Neighbor Discovery as the Layer 2 destination address.

- j. Select the first **NDP** event at SwitchA.

Is there any difference between the In Layers and Out Layers for Layer 2?

No. The switch only sends the frame on; it doesn't change Layer 2 information.

- k. Select the first **NDP** event at **PCA2**. Click the Outbound PDU Details.

What addresses are displayed for the following?

Note: The addresses in the fields may be wrapped, adjust the size of the PDU window to make address information easier to read.

Ethernet II DEST ADDR:

ADDR: 0001.427E.E8ED

Ethernet II SRC ADDR:

ADDR: 0040.0B02::243E

IPv6 SRC IP:

2001:db8:acad:1::b

IPv6 DST IP:

: 2001:db8:acad:1::a

- l. Select the first **NDP** event at **RTA**. Why are there no Out Layers?

The packet is dropped because the IPv6 address does not match the router's address.

- m. Click through the **Next Layer >>** button until the end and read steps 4 through 7 for further explanation.
- n. Click the next **ICMPv6** event at **PCA1**.

Does PCA1 now have all of the necessary information to communicate with PCA2?

Yes, it now knows both the IPv6 address and the MAC address of PCA2's destination.

- o. Click the last **ICMPv6** event at **PCA1**. Notice this is the last communication listed.

What is the ICMPv6 Echo Message Type?

The ICMPv6 Echo Message Type is 129, an echo reply.

- p. Click the **Reset Simulation** button in the Simulation Panel. From the command prompt of PCA1 repeat the **ping** to PCA2. (Hint: you should be able to press the up arrow to bring the previous command back.)
- q. Click the **Capture Forward** button 5 times to complete the ping process.

Why weren't there any NDP events?

PCA1 doesn't need to use Neighbor Discovery because it already knows PCA2's MAC address.

Part 2: IPv6 Neighbor Discovery Remote Network

In Part 2 of this activity, you will perform steps that are similar to those in Part 1, except in this case, the destination host is on another LAN. Observe how the Neighbor Discovery process differs from the process you observed in Part 1. Pay close attention to some of the additional addressing steps that take place when a device communicates with a device that is on a different network.

Make sure to click the **Reset Simulation** button to clear out the previous events.

Step 1: Capture events for remote communication.

- a. Display and clear any entries in the IPv6 neighbor device table as was done in Part I.
- b. Switch to simulation mode. Click the **Show All/None** button in the lower left part of the Simulation Panel. Make certain the **Event List Filters – Visible Events** displays **None**.
- c. From the command prompt on PCA1 issue the command **ping -n 1 2001:db8:acad:2::a** to ping host PCB1.
- d. Click the **Play Capture Forward** button which is displayed as an arrow pointing to the right with a vertical bar within the Play Controls box. The status bar above the Play Controls should read Captured to 150. (The exact number may vary.)
- e. Click the **Edit Filters** button. Select the IPv6 tab at the top and check the boxes for **ICMPv6** and **NDP**. Click the red X in the upper right of the Edit ACL Filters window. All of the previous events should now be listed. You should notice there are considerably more entries listed this time.
- f. Click the square in the Type Column for the first event, which should be **ICMPv6**. Because the message starts with this event, there is only an Outbound PDU. Notice that it is missing the Layer 2 information as it did in the previous scenario.
- g. Click the first **NDP** event At Device **PCA1**.

What address is being used for the Src IP in the inbound PDU?

The Link Local address for PCA1 – fe80::201:42ff:fe7e:e8ed

IPv6 Neighbor Discovery will determine the next destination to forward the ICMPv6 message.

- h. Click the second ICMPv6 event for **PCA1**. PCA1 now has enough information to create an ICMPv6 echo request.

What MAC address is being used for the destination MAC?

0001.961d.6301, the MAC address of G0/0/0 of RTA

- i. Click the next ICMPv6 event at device **RTA**. Notice that the outbound PDU from RTA lacks the destination Layer 2 address. This means that RTA once again has to perform a Neighbor Discovery for the interface that has the 2001:db8:acad:2:: network because it doesn't know the MAC addresses of the devices on the G0/0/1 LAN.

- j. Skip down to the first ICMPv6 event for device **PCB1**.

What is missing in the outbound Layer 2 information?

The destination MAC address must be determined for the IPv6 destination address.

- k. The next few **NDP** events are associating the remaining IPv6 addresses to MAC addresses. The previous NDP events associated MAC addresses with Link Local addresses.
- l. Skip to the last set of ICMPv6 events and notice that all of the addresses have been learned. The required information is now known, so PCB1 can send echo reply messages to PCA1.
- m. Click the Reset Simulation button in the Simulation Panel. From the command prompt of PCA1 repeat the command to ping PCB1.
- n. Click the Capture Forward button nine times to complete the ping process.

Were there any NDP events?

No

- o. Click the only **PCB1** event in the new list.

What does the destination MAC address correspond to?

The router interface G0/0/1 on RTA

Why is PCB1 using the router interface MAC address to make its ICMP PDUs?

Because the destination device is on another network, PCB1 addresses the PDU to the default gateway interface MAC. RTA will determine how to address the PDU at Layer 2 to send it towards its destination.

Step 2: Examine router outputs.

- a. Return to **Realtime** mode.
- b. Click **RTA** and select the CLI tab. At the router prompt enter the command **show ipv6 neighbors**.

How many addresses are listed?

4 – IPv6 global unicast and link local addresses and MAC addresses for PCA1 and PCB1

What devices are these addresses associated with?

PCA1 and PCB1

Are there any entries for PCA2 listed (why or why not)?

PCA2 has not communicated across the network yet.

Ping **PCA2** from the router.

- c. Issue the **show ipv6 neighbors** command.

Are there entries for PCA2?

Yes, the IPv6 address and MAC address for PCA2

Reflection Questions

1. When does a device require the IPv6 Neighbor Discovery process?

When the destination MAC address is not known. This process is similar to ARP with IPv4.

2. How does a router help to minimize the amount of IPv6 Neighbor Discovery traffic on a network?

The router keeps neighbor tables so that it doesn't have to start ND for every host it needs to reach.

How does IPv6 minimize the impact of the ND process on network hosts?

It uses a multicast address so only a small number of addresses receive the Neighbor Discovery messages. IPv6 generates a specially designed multicast destination MAC address that contains part of the node's address.

3. How does the Neighbor Discovery process differ when a destination host is on the same LAN and when it is on a remote LAN?

If the destination host is on the same LAN segment, only the device with the matching IPv6 address replies, while all other devices ignore the packet. When the destination is on a different network, the gateway device (usually a router) supplies the MAC address of its local interface as the destination MAC, then looks for the correct MAC address on the remote network. The router then stores the responding IPv6 and MAC address pair in the IPv6 Neighbor table, similar to an ARP table in IPv4.